

Final-year Bachelor of Science in Information Technology student with hands-on experience building full-stack web applications, AI/ML pipelines, and deep learning systems. Proficient in Python, JavaScript, and modern frameworks with a strong foundation in RAG systems, multi-sensor deep learning, RESTful APIs, and scalable architecture. Seeking a Software Developer or AI/ML Internship to contribute to production-level applications and collaborative engineering teams.

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, C#, PHP, SQL
Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS
Backend: Node.js, Express.js, RESTful APIs, ASP.NET MVC
Databases: MongoDB, SQL Server, MySQL
AI / Systems: PyTorch, SentenceTransformers, FAISS, Ollama, RAG Pipelines, Transformer, BiLSTM, CNN, Vertex AI, Google Gemini, Multi-Agent Systems
Databases: MongoDB, SQL Server, MySQL, Vector Databases (FAISS)
Tools: Git, Linux, GitHub, VS Code, Visual Studio

EDUCATION

Bachelor Of Science in Information Technology 2022-2026
Fairleigh Dickinson University, Vancouver, BC

- Received Academic Honors (Spring 2025)
- Relevant Coursework: Full-Stack Development, OOP (Java/C#), Web Development, Data communication

PROJECTS

BFRB Gesture Recognition- using Deep Learning & NLP *Python, PyTorch, CNN, BiLSTM, Transformers, scikit-learn, Pandas, NumPy*

- Led development of an 18-class gesture classifier detecting Body-Focused Repetitive Behaviors (BFRBs) using wearable IMU, thermopile, and time-of-flight sensor data.
- Implemented and evaluated 8 model architectures including CNN-BiLSTM, Late Fusion Ensemble, FFT-MLP, and NLP-based novels Transformers and BiLSTM models.
- NLP Transformer model achieved 85.98% macro F1, outperforming all paper baselines by 35+ points; Late Fusion model reached 94.28% binary F1 for BFRB detection.
- Designed reproducible ML pipeline with fixed random seeds, sequence-level train/test splits, and serialized artifacts for consistent evaluation across all models.

Local RAG System Prototype — Fully Offline AI Knowledge Base *Python, SentenceTransformers, FAISS, Ollama, LLM (LLaMA 3.2)*

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline running entirely locally with no external API calls, covering ingestion, embedding, vector search, and grounded answer generation.
- Used SentenceTransformers (all-MiniLM-L6-v2) for embedding and FAISS for vector indexing, with Ollama to serve a local LLaMA model for context-grounded generation.
- Implemented hallucination control with intentionally fake training data, strict prompt constraints, and source citations — ensuring all answers come from retrieved context.
- Architected modular Python codebase with separate ingest, embed, retrieval, and generation modules, designed for extension with PDF ingestion, reranking, and FastAPI interface.

AI Multi-Agent System *Python, Flask, Google Gemini 1.0 Pro, Vertex AI*

- Designed and implemented a 5-agent AI system for personalized Python tutoring, with specialized agents handling teaching, practice generation, progress tracking, feedback, and guidance.
- Integrated Google Gemini 1.0 Pro models to power interactive teaching and adaptive practice modules, enabling intelligent and context-aware student responses.
- Architected system for deployment on Vertex AI Agent Engine, supporting scalable, cloud-ready multi-agent orchestration.

Network Packet Analyzer (2025) *Python, Scapy, Rich, React*

- Built an educational packet analyzer capturing and parsing Ethernet, IP, TCP/UDP, and ICMP packets with real-time anomaly detection and protocol-level statistics.
- Delivered an interactive CLI tool with safe network simulations, filtering, and visualizations for both professional diagnostics and learning purposes.

Full-Stack Portfolio Website (2025) *React.js, Tailwind CSS, Node.js, Express.js, MongoDB*

- Developed a production-deployed personal portfolio with dark/light mode, smooth animations, and reusable React components.
- Built backend REST APIs with Node.js/Express connected to MongoDB for contact form handling and data persistence.

Travel Booking System (2024–2025) — ASP.NET MVC, C#, SQL Server, Bootstrap

- Developed a full-stack booking platform with user authentication, CRUD operations, and automated email notifications.
- Architected normalized SQL Server schema for users, destinations, and reservations with a responsive Bootstrap UI.

VOLUNTEER EXPERIENCE

UN Youth Delegate *United Nations, New York, NY | May 2025*

- Represented youth perspectives in global dialogue, developing advanced communication and cross-cultural leadership skills.

Event Coordinator *Fairleigh Dickinson University – Vancouver, BC | Mar 2024 – Present*

- Coordinated campus events, strengthening teamwork, logistics, and problem-solving skills in fast-paced environments.